Data Communication & Computer Networks (CN)

Topic 7 of 7 : Network Security

47 Pending Questions — staged from repo exam PDFs & trusted sources — **NEED VERIFICATION** (answers + Hindi to be verified during topic build)

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)**

Total Questions: **47** pending questions (staging)

Question Type: MCQ — 4 options (A, B, C, D) per question, in English + Hindi

Layout: Left = **question** | Right = **explanation** (green): ✔ **why the correct answer is right** + ★ **Nukta** (key exam point)

Language rule: Hindi is the second language; **technical terms stay in English** (topology, protocol, jitter…)

Verification: Every answer cross-verified **4×** (Forouzan + official answer keys of Subject Test / Unit Test Solutions). Wrong options verified internally.

Syllabus Coverage — sub-topic wise order

How to read: Left side: question + 4 options (try it yourself first). Right side (green): ✔ correct answer with why it is right, and ★ **Nukta** — the one-line exam tip to remember. Practice in 2 passes: first read question + options only, then check the green column.

System: generated by **tools/cn_build_topic.py** from **tools/data/topic7_data.py**. To add more questions later: drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions auto-classify topic-wise. See **Computer Network/README.md**.

Pending Review — auto-ingested, needs verification

Auto-extracted & classified; answers to be verified before counting.

P1. Network Attacks & Description 9 List–I (Attack
(A) Phishing ( )
(B) DoS Attack
(C) Man-in-the-Middle
(D) Spoofing (R) List–II (Description /) (i) Flooding network traffic / 9 (ii) Stealing user credentials (iii) Intercepting communication B W+ (iv) Impersonation attack Codes / : (A) a–ii, b–i, c–iii, d–iv (B) a–i, b–ii, c–iii, d–iv (C) a–iii, b–i, c–ii, d–iv (D) a–ii, b–iii, c–i, d–iv

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

P2. In the digital signature technique when the whole message is signed using an asymmetric key, the receiver of the message uses _____ to verify the signature. v B Asymmetric Key W + , - y W + z _____

(A) Her or his own public key
(B) Her or his own private key
(C) The sender’s public key
(D) None of these

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P3. The text that gets transformed using an algorithm cipher is called: " $ 2 $: ;

(A) Complex Text
(B) Transformed Text
(C) Plain Text
(D) Scalar Text

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P4. In cryptography, which statement about Plaintext, Ciphertext and Key is MOST accurate? ‰Š ‹ B \ Œ, Œ

(A) Plaintext is generated after decryption using a public key
(B) Ciphertext is the original readable message
(C) Plaintext is transformed into Ciphertext using an algorithm and a key
(D) Ciphertext and plaintext are always of the same length

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P5. Which attack becomes possible if the same key is reused multiple times in symmetric encryption

(A) Replay Attack
(B) Known Plaintext Attack
(C) Man-in-the-Middle Attack
(D) Phishing Attack

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

P6. Data Encryption Standard (DES) ; ?Data Encryption Standard (DES) is a:

(A) Block Cipher
(B) Stream Cipher
(C) Bit Cipher
(D) Byte Cipher

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P7. In the virus life cycle, which phase actually performs the malicious activity? H B - h

(A) Infection Phase / Z
(B) Triggering Phase
(C) Dormant Phase / C
(D) Execution (Payload) Phase / F

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

P8. Which of the following backup media provides Write Once Read Many (WORM) functionality? k l B Write Once Read Many (WORM) @

(A) Hard Disk / h 1
(B) Magnetic Tape / j
(C) Optical Disk (CD-R / DVD-R)
(D) Flash Drive / k 9

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

P9. IPSec operates in which mode(s)? IPSec B P

(A) Transport mode only / 9 h
(B) Tunnel mode only
(C) Transport mode and Tunnel mode / 9 h
(D) Application mode / n$

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IPSec & Security Protocols]

⚠ auto-ingested — answer needs verification

P10. SSL/TLS handshake protocol is mainly used for:SSL/TLS u @ v w x ;

(A) Data encryption
(B) Data compression
(C) Key exchange and authentication
(D) Packet routing

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IPSec & Security Protocols]

⚠ auto-ingested — answer needs verification

P11. IPSec is MOST commonly used to implement: IPSec

(A) Secure Email
(B) Secure Web Browsing
(C) Virtual Private Network (VPN)
(D) Wireless Security

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

P12. In IPSec Tunnel mode:/ IPSec B ;

(A) Only payload is encrypted
(B) Only header is encrypted
(C) Entire IP packet is encrypted
(D) No encryption is used

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IPSec & Security Protocols]

⚠ auto-ingested — answer needs verification

P13. Which of the following uses the same key for encryption and decryption? H I + - I}y~ y~

(A) Symmetric Key Cipher
(B) Asymmetric Key Cipher
(C) Hash Function
(D) Digital Signature

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P14. Which of the following uses a pair of keys (public and private)? H I + (9)

(A) Symmetric Cipher
(B) Asymmetric Cipher
(C) DES
(D) AES

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P15. Which of the following is faster in encryption and decryption? I}y~ y~ + -

(A) Symmetric Key Cipher
(B) Asymmetric Key Cipher
(C) Can not decide
(D) None of the these

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P16. Public key is used to _____. I• _____

(A) Encrypt the data
(B) Decrypt the data
(C) Both encrypt and decrypt
(D) Generate session keys

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P17. Which of the following is NOT a symmetric encryption algorithm? H I I}y~ € d •

(A) AES
(B) DES
(C) Blowfish
(D) RSA

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P18. What is the main disadvantage of symmetric key encryption? h I}y~ T

(A) It is very slow
(B) Key distribution is difficult
(C) It uses two keys
(D) It cannot encrypt text

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P19. In public key cryptography, the message encrypted with the public key can only be decrypted by _____. I• y, m +, I• I}y, _____ ƒ

(A) Another public key
(B) Same public key
(C) Private key corresponding to that public key
(D) Session key

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P20. Which one of the following statements is true about symmetric and asymmetric cryptography? h h y, m + H + -

(A) Symmetric cryptography provides authentication.
(B) Asymmetric cryptography is faster than symmetri
(C) C. Symmetric cryptography requires secure key exchange.
(D) Both use two different keys.

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P21. What is the key length of AES-256 encryption? AES-256 I}y~

(A) 128 bits
(B) 192 bits
(C) 256 bits
(D) 512 bits

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P22. Which cipher method encrypts one bit or byte at a time? I}y,

(A) Block Cipher
(B) Stream Cipher
(C) Substitution Cipher
(D) Transposition Cipher

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P23. Which of the following statements is FALSE? H + -

(A) AES is a block cipher.
(B) RSA is a symmetric cipher.
(C) Diffie-Hellman is used for key exchange.
(D) DES is symmetric. Which of the following is NOT a goal of network security? H I + - 9 t „ ? A. Confidentiality ( ) B. Integrity ( ) C. Availability (Y) D. Redundancy ( )

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P24. The term "CIA triad" in information security refers to _____. t + "CIA] " _____
(A) Cryptography, Integrity, Authentication
(B) Confidentiality, Integrity, Availability
(C) Confidentiality, Information, Access
(D) Control, Information, Access
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P25. The goal of “Integrity” ensures that
(A) Data can be accessed only by authorized users
(B) Data is accurate and has not been modified
(C) Data is always available when needed
(D) Unauthorized users cannot enter the system 40 Which of the following attack violates confidentiality? H I + j ? A. Interception B. Modification C. Fabrication D. Interruption
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P26. Modification attack violates which security goal? t „ j
(A) Confidentiality
(B) Integrity
(C) Availability
(D) Authentication
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P27. Denial of Service (DoS) attack violates which goal? Denial of Service (DoS) „ j
(A) Confidentiality
(B) Integrity
(C) Availability
(D) Authentication
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

P28. Spoofing attack targets which type of security aspect? ˆ t t
(A) Authentication
(B) Integrity
(C) Availability
(D) Confidentiality
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P29. Which of the following is an active attack? H I + - y (Active)
(A) Interception
(B) Modification
(C) Traffic Analysis
(D) Eavesdropping
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

P30. The Caesar Cipher is an example of which type of cryptographic technique? Caesar Cipher y, m

(A) Transposition Cipher
(B) Substitution Cipher
(C) Stream Cipher
(D) Block Cipher In Steganography, the file used to hide secret data is called _____. Steganography + + p A. Secret File B. Cover File C. Cipher File D. Key File

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P31. The process of detecting hidden data inside a file is known as _____. ? y

(A) Cryptanalysis
(B) Steganalysis
(C) Data Mining
(D) Stegnography

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P32. What is the output of a Steganography process called? Steganography ? y

(A) Ciphertext
(B) Stego-object
(C) Encoded file
(D) Key file

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 7 / Cryptography]

⚠ auto-ingested — answer needs verification

P33. Which of the following is used in HTTPS for encryption?HTTPS

(A) SSL/TLS
(B) FTP
(C) ARP
(D) NAT

[Source: CN Topic Test 5.pdf | auto-classified: Topic 7 / IPSec & Security Protocols]

⚠ auto-ingested — answer needs verification

P34. What type of NAT maps multiple private IPs to one public IP? NAT IP IP

(A) Static NAT
(B) Dynamic NAT
(C) PAT (Port Address Translation)
(D) Double NAT

[Source: CN Topic Test 5.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

P35. Which mode of IPsec provides end-to- end security?IPsec

(A) Transport mode
(B) Tunnel mode
(C) Static mode
(D) Gateway mode What type of attack is a Smurf attack?Smurf (a) DoS (b) Spoofing (c) Phishing (d) Brute force

[Source: CN Topic Test 5.pdf | auto-classified: Topic 7 / IDS / IPS / Firewall]

⚠ auto-ingested — answer needs verification

P36. What is the function of a proxy server
(A) Asssign IP address
(B) Direct connect to the internet
(C) Act as an intermediary between client and server
(D) Block all network traffic

[Source: CN Topic Test 5.pdf | auto-classified: Topic 1 / Types of Networks]

⚠ auto-ingested — answer needs verification

P37. What is the advantage of setting up a DMZ (Demilitarized Zone) using firewall?
(A) To create multiple zones of trust
(B) Load balancing on the server
(C) Improve network performance
(D) Brings internet service into home from internet service provider

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 1 / Types of Networks]

⚠ auto-ingested — answer needs verification

P38. Match the following Security Protocols with their corresponding OSI Layers: # % + - 2 OSI 9 : List–I (Protocol) List–II (Layer)
(A) IPSec (i) Application Layer
(B) SSL / TLS (ii) Network Layer
(C) PGP (Pretty Good Privacy) (iii) Transport / Presentation Layer
(D) WPA2 (iv) Data Link Layer (Wi-Fi) Choose the correct answer from the codes given below: (A) a–ii, b–iii, c–i, d–iv (B) a–ii, b–i, c–iii, d–iv (C) a–iii, b–ii, c–i, d–iv (D) a–i, b–ii, c–iii, d–iv

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P39. Security Mechanisms & OSI Layers + X OSI B List–I (Mechanism / X)
(A) Firewall ( )
(B) Digital Signature (W +)
(C) MAC Filtering (MAC YZ)
(D) Port-based Filtering (9 Z YZ) List–II (Layer /) (i) Network Layer / 9 (ii) Application Layer / %\ (iii) Data Link Layer (iv) Transport Layer / S 9 Codes / : (A) a–i, b–ii, c–iii, d–iv (B) a–ii, b–i, c–iii, d–iv (C) a–iv, b–ii, c–i, d–iii (D) a–i, b–iii, c–ii, d–iv

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P40. The primary difference between active and passive attacks is _____. Active Passive + T
(A) Active attacks change data; passive do not.
(B) Passive attacks destroy hardware.
(C) Both are same.
(D) Passive attacks are legal.

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 2 / OSI vs TCP/IP & Layer Functions]

⚠ auto-ingested — answer needs verification

P41. Which of the following is an example of asymmetric encryption? **(A)** AES **(B)** DES **(C)** RSA **(D)** Blowfish [Source: SSC CGL Tier-2 repeated Qs (via Oliveboard) | auto-classified: Topic 7 / Cryptography]	⚠ Site key.
P42. Nelwonk seuniy Which of the following is NOT a goal of network security? goads. **(A)** Confidentiality (Jlu) **(B)** Integrity (3sI) **(C)** Availability (3uvcT) Gonldatiaity . Redundancy (3fqyeI) ompuTechAcademy ANINSTITUTEFOR Integily Aralakility [Source: Subject Test Solution -1 (CN & Security) Part-2.pdf (OCR), p12 | auto-classified: Topic 7 / Cryptography]	⚠ OCR-parse staging: verify options/answer at curation (see src page).
P43. SUNILYADAVSIR Which of the following attacks violates confidentiality? pmvauw Interception **(A)** Modification **(B)** Fabrication CompuTechp. Interruption AN INSTITUTEFOR [Source: Subject Test Solution -1 (CN & Security) Part-2.pdf (OCR), p15 | auto-classified: Topic 7 / Cryptography]	⚠ OCR-parse staging: verify options/answer at curation (see src page).
P44. In Steganography, the file used to hide secret data is called Steganography H u **(A)** Secret File Cover File **(B)** Cipher File chAcademy **(C)** Key File AN INSTITUTEFOR [Source: Subject Test Solution -1 (CN & Security) Part-2.pdf (OCR), p23 | auto-classified: Topic 7 / Cryptography]	⚠ OCR-parse staging: verify options/answer at curation (see src page).
P45. 22. The types of Medium Access s Control are **(A)** Centrally Access Control **(B)** Distributed Access Control **(C)** both a and b **(D)** None of the above [Source: Unit Test -2 Solutions.pdf (OCR), p24 | auto-classified: Topic 7 / IDS / IPS / Firewall]	⚠ OCR-parse staging: verify options/answer at curation (see src page).
P46. 5. In which layer of network architecture, the secured socket layer (SsL) is used? becwu QuyeeTL ≤ .Socke **(A)** Physical layer **(B)** Session layer bayen Appli ah'on **(C)** Application layer y Presentation layer Toanrpor byer [Source: Unit Test-4 Solution.pdf (OCR), p6 | auto-classified: Topic 7 / OSI Model Basics]	⚠ SSL layer mapping — exam banks usually key 'Session layer'; verify at curation (source p6).

P47. Which of the following port and IP address scanner is famous among users? # % B - 9 IP S „ 9 - †

(A) Cain and Abel
(B) Angry IP Scanner
(C) Snort
(D) Ettercap

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 7 / IPv4 Addressing]

⚠ moved from Topic 3 during T3 curation (belongs to Topic 7)